

Prakhar Shekhar Parthasarthi

Full-Stack Software Engineer · Java & Spring Boot · React · AWS

prakhar.mnnit.2022@gmail.com

Shahjahanpur, UP · open to relocate

github.com/praxstack | linkedin.com/in/prakharshekhar | prax-portfolio-one.vercel.app

47%

p90 latency cut

+33%

conversion lift

195k

GET rps · Java Redis

20 → 5

on-call / shift

SUMMARY

Full-stack software engineer with **3+ years at Amazon** across Payments and Travel — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture`, bounded `ExecutorService`, `ConcurrentHashMap`), idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure in **AWS CDK (Java)** — paired with **React (SSR & CSR)** + **Redux** for customer-facing flows.

SKILLS

Java & JVM	Java 17, Spring Boot, Spring MVC, Guice, JUnit 5, Mockito, SLF4J, Maven, Gradle · <code>ExecutorService</code> , <code>CompletableFuture</code> , <code>ConcurrentHashMap</code> , JVM profiling, GC analysis
Distributed Systems	Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry/backoff, DLQ, at-least-once, TDD
AWS	Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java)
Front-end	React.js (SSR & CSR), Redux, HTML/CSS, JavaScript/TypeScript
Data & Storage	DynamoDB, Redis, NoSQL modelling, SQL, caching strategies
Tooling & Process	Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile/Scrum, on-call, code review, incident response · Python, C++

EXPERIENCE

Software Development Engineer, Jul 2022 — Sep 2025

Amazon · Travel (Hotels & Flights) · Bengaluru

- Service ownership — Spring Boot, Spring MVC. Owned **4 customer-facing microservices** powering the hotel booking platform — search, listing, room selection, order review — across design, code review, deploy, and on-call. REST controllers, service-layer logic, `@Repository` DynamoDB DAOs via Spring DI. Prime-exclusive discount and partner-discount integration — ~30 bps funnel churn drop.
- JVM concurrency — `CompletableFuture`, bounded executors. Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 cut 47% (1.1s → 0.59s), conversion up +33%. GZIP on 150–200 MB catalog payloads cut wire size 95%.
- Event-driven — SQS + Step Functions. Designed idempotent Java webhook for real-time flight schedule changes — SQS ingestion, DynamoDB dedup keys, audit logging, automated offline modifications — 43 bps CPT. Orchestrated booking state machine (confirm/cancel/retry) on Step Functions with Java Lambda handlers.
- Infrastructure as code — AWS CDK (Java). Authored all service IaC in **Java CDK** — Lambdas, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch alarms. Migrated 2 services from hand-rolled CloudFormation — rollback incidents halved.

Jul 2022 — Sep 2025 · Travel (Hotels & Flights) · Bengaluru

- **Testing** — **JUnit 5**, **Mockito**, **Spring Boot**. Held **80%+** line coverage; Mockito argument captors for downstream contract verification; `@SpringBootTest` integration tests against in-memory DynamoDB. Introduced Gradle JaCoCo coverage gates in CI.
- **Operational excellence**. On-call tickets cut from **20+** to **~5** per shift (**~2** SDE-weeks/week reclaimed). Java email-automation deprecated stale WebLab experiments — config-hygiene incidents from **5–6** to **0**. Resolved **150+** production incidents; authored COEs. Built auto-insurance purchase flow (Java + React) — **60%+** renewal lift.

Software Development Engineer, Intern, May 2021 – Jul 2021 · Feb 2022 – Jul 2022

Amazon · Bengaluru

- Migrated legacy GraphQL resolvers to **Java** — DynamoDB via Spring-injected DAOs, API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance using past-purchase data — async Java pre-fill workflow.
- Applied OO design and **Google Guice DI**; **90%+** unit-test coverage (JUnit 5 + Mockito).

SELECTED PROJECTS

AI Visual Code Review github.com/praxstack/ai-visual-code-review · npm <code>Node.js</code> <code>TypeScript</code> <code>Jest</code> <code>VS Code API</code> Security-hardened visual code-review tool with GitHub-style diff UI, AI markdown exports, command-injection protection, rate limiting. Published npm + VS Code extension. Published: npm v2.4.1 · MIT.	Redis Server in Java github.com/praxstack/redis-server-java <code>Java 17</code> <code>ConcurrentHashMap</code> <code>JUnit 5</code> RESP2-compatible Redis from scratch; bounded <code>ExecutorService</code>, lock-free reads, hybrid TTL eviction, TCP integration tests. GET 195k rps · SET 154k rps · p50 < 0.3 ms · 42 tests.
Audio Transcription Pipeline github.com/praxstack/audio-transcription-pipeline <code>Python</code> <code>Bedrock</code> <code>pyannote</code> <code>pytest</code> Multi-stage pipeline: raw audio/video → speaker-labelled Markdown. pyannote diarization + AWS Bedrock (Voxtral ASR + Claude repair). 63 tests · 6 stages · MIT.	Coach Atlas github.com/praxstack/coach-atlas <code>React</code> <code>TypeScript</code> <code>Mermaid</code> <code>KaTeX</code> AI technical-interview mentor + tutorial studio; Socratic prep, Mermaid/KaTeX/code rendering, multi-provider BYOK (keys in browser). React + TypeScript · deployed on Vercel.

Open-Source Contributions

Merged into danielmiessler/Fabric (Go, 42k stars):
#2044 — AWS Bedrock bearer-token (ABSK) auth + guided `--setup` . +882 / -83
#2052 — dynamic Bedrock region fetching (40+ regions). +252 / -15
#2061 — streaming-goroutine deadlock fix in `Chatter.Send` . +494 / -42
Under review: deer-flow, claude-mem, tolaria.

EDUCATION

Master of Computer Applications (M.C.A.), Computer Science · MNNIT, Prayagraj · GPA 9.25 · 2019 — 2022
Bachelor of Science (B.Sc. Hons), Computer Science · Bhaskaracharya College, University of Delhi · 2015 — 2018

AWARDS

Amazon Pay Merchants Categories Champion — 100+ hotel trust-buster bugs; on-call 20+ to ~5 per shift · 2023